Typical Environmental Overlaps in High-Dimensional Hilbert Space: Kinematic Bounds for Decoherence

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We quantify a conditional kinematic contribution to environmental decoherence. For independent Haar-random pure states in a D-dimensional complex Hilbert space, the squared overlap has distribution $\mathrm{Beta}(1, D-1)$, and therefore $\mathbb{P}(|\langle\phi|\psi\rangle|^2 \geq \varepsilon) = (1-\varepsilon)^{D-1}$. We combine this identity with a union bound to control the trace distance between the reduced state of a finite family of pointer alternatives and its completely dephased counterpart; a complementary Hilbert–Schmidt estimate controls the collective contribution of many weak coherences. For a mixed initial environment and a Haar-random relative conditional unitary W, we obtain $\mathbb{E}|\operatorname{Tr}(\rho_E W)|^2 = \operatorname{Tr}(\rho_E^2)/D$ and convert this identity into simultaneous high-probability bounds for a finite branch family. These statements are conditional on suitable typicality of the relative environmental dynamics. High dimension alone neither establishes such typicality nor selects a pointer basis. Moreover, suppression of system coherence must be distinguished from the formation of distinguishable or redundantly accessible environmental records. The analysis thus isolates the geometric capacity used by decoherence while making explicit the dynamical and information-theoretic assumptions required for its physical application.

I. INTRODUCTION

The quantum measurement problem includes several logically distinct questions: which observables acquire a preferred status, why individual measurements have definite outcomes, and why their statistics follow the Born rule [1, 2]. Environmental decoherence addresses an important operational part of this problem. It explains why interference between certain alternatives becomes inaccessible to measurements on a system once information about those alternatives has dispersed into its environment. Decoherence does not, by itself, select one member of the resulting global superposition as the unique outcome.

Beyond his famous cat thought experiment introduced seventeen years prior [3], Schrödinger dramatized the tension in a 1952 Dublin colloquium. If the wave equation applies universally, he observed, a naive reading appears to turn macroscopic surroundings into a “quagmire” or “featureless jelly” and even seems to threaten the stability of an unobserved wristwatch in a drawer [4].

The modern operational response is not that the global superposition disappears. Rather, environmental correlations make its relative phases inaccessible to the restricted observables by which macroscopic systems are ordinarily interrogated.

Standard decoherence theory supplies the dynamical framework: the system–environment interaction selects stable pointer observables and correlates their alternatives with conditional environmental states [1, 5–7]. High dimension alone does not guarantee decoherence: a relative conditional unitary may, for example, remain the identity on the entire environmental space. The question addressed here is narrower: once an explicit typicality assumption for the conditional environmental states or relative unitaries has been specified, what suppression follows from high-dimensional geometry alone? Unit vectors sampled independently from a high-dimensional complex Hilbert space normally have very small overlaps. This is an elementary instance of concentration of measure [8–11].

The word *independently* is essential. Actual environmental branch states arise from the same initial state under correlated conditional dynamics. Their relative unitary need not be Haar distributed, and a large energy shell does not make it so. We therefore formulate the geometric results as conditional bounds, identify the precise dynamical object to which they would apply, and distinguish three notions that are easily conflated: small reduced-system coherences, distinguishable conditional environmental states, and redundant classical records.

The individual geometric identities used below are standard. The purpose of this paper is to combine them into operational trace-distance estimates for finite families of conditional environmental states, to extend the mean-square analysis to mixed environments, and to delineate the dynamical and information-theoretic assumptions needed before these kinematic estimates can be interpreted as physical decoherence claims.

II. DECOHERENCE FACTORS AND AN OPERATIONAL BOUND

Let $\{|s_i\rangle\}_{i=1}^m$ be orthonormal pointer alternatives spanning the support of the system state, selected by a measurement-like interaction, and let $\sum_i |c_i|^2 = 1$. For a pure initial environment $|E_0\rangle$, the joint evolution has the

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form $|E_i(t)\rangle = U_i(t)|E_0\rangle$, with

$$\left(\sum_{i=1}^{m} c_i|s_i\rangle\right)|E_0\rangle \longmapsto |\Psi(t)\rangle = \sum_{i=1}^{m} c_i|s_i\rangle|E_i(t)\rangle. \tag{1}$$

Tracing out the environment gives

$$\rho_S(t) = \sum_{i,j=1}^{m} c_i c_j^* \gamma_{ji}(t)|s_i\rangle\langle s_j|, \qquad \gamma_{ji}(t) = \langle E_j(t)|E_i(t)\rangle. \tag{2}$$

Let Δ denote complete dephasing in the pointer basis,

$$\Delta(\rho_S) = \sum_i |s_i\rangle\langle s_i|\rho_S|s_i\rangle\langle s_i|. \tag{3}$$

Thus Δ is trace preserving on the relevant support. All coherence statements below are relative to this dynamically selected pointer basis; they do not define a basis-independent quantity called coherence. The trace distance $D_{\mathrm{tr}}(\rho,\sigma) = \frac{1}{2}\|\rho-\sigma\|_1$ has a direct operational meaning: it controls the maximum change of outcome probabilities over all measurements. The environmental overlaps give the following elementary bound.

Proposition 1 (Reduced-state coherence bound)**.** *For the state in Eq.* (2),

$$D_{\mathrm{tr}}(\rho_S, \Delta(\rho_S)) \le \min\left\{1, \frac{1}{2}\sum_{i\neq j} |c_i c_j|\,|\gamma_{ji}|\right\}. \tag{4}$$

It also obeys the Hilbert–Schmidt bound

$$D_{\mathrm{tr}}(\rho_S, \Delta(\rho_S)) \le \min\left\{1, \frac{1}{2}\sqrt{mZ}\right\}, \tag{5}$$

where

$$Z = \sum_{i\neq j} |c_i|^2|c_j|^2|\gamma_{ji}|^2. \tag{6}$$

If $|\gamma_{ji}| \le \delta$ for all $i \neq j$, then

$$D_{\mathrm{tr}}(\rho_S, \Delta(\rho_S)) \le \min\left\{1, \frac{\delta}{2}\left[\left(\sum_i |c_i|\right)^2 - 1\right]\right\} \le \min\left\{1, \frac{m-1}{2}\delta\right\}. \tag{7}$$

For two alternatives the first inequality is an equality: $D_{\mathrm{tr}} = |c_1 c_2|\,|\gamma_{12}|$.

Proof. Subtracting the dephased state leaves $\sum_{i\neq j} c_i c_j^* \gamma_{ji}|s_i\rangle\langle s_j|$. The triangle inequality, the trivial bound $D_{\mathrm{tr}} \le 1$, and $\||s_i\rangle\langle s_j|\|_1 = 1$ give Eq. (4). Since the difference has rank at most m, $\|X\|_1 \le \sqrt{m}\|X\|_2$ and direct evaluation of $\|X\|_2^2 = \mathrm{Tr}(X^\dagger X)$ give Eq. (5). The first bound in Eq. (7) follows by taking the maximum overlap out of the sum. The second follows from $\sum_i |c_i| \le \sqrt{m}$ and $\sum_i |c_i|^2 = 1$. For $m = 2$, the two nonzero singular values of the off-diagonal difference are both $|c_1 c_2 \gamma_{12}|$. □

Pairwise quasi-orthogonality is therefore useful only together with control of the number and amplitudes of the alternatives. A tiny entrywise bound can fail to control a large Gram matrix if sufficiently many off-diagonal entries accumulate.

III. HAAR GEOMETRY AND SIMULTANEOUS BRANCH BOUNDS

A. Exact overlap distribution

Lemma 1 (Complex Haar overlap)**.** *Let $|\psi\rangle, |\phi\rangle \in \mathbb{C}^D$, $D \ge 2$, be independent Haar-random unit vectors. Then*

$$T = |\langle\phi|\psi\rangle|^2 \sim \mathrm{Beta}(1, D-1), \tag{8}$$

and, for $0 \le \varepsilon \le 1$,

$$\mathbb{P}(T \ge \varepsilon) = (1-\varepsilon)^{D-1} \le e^{-(D-1)\varepsilon}. \tag{9}$$

In particular, $\mathbb{E}T = 1/D$.

Proof. By unitary invariance fix $|\phi\rangle = |e_1\rangle$. A Haar-random unit vector can be written as $Z/\|Z\|$, where the Z_k are independent standard complex Gaussian variables. The variables $X_k = |Z_k|^2$ are independent exponentials of mean one. Hence

$$T = \frac{X_1}{X_1 + \cdots + X_D} \sim \mathrm{Beta}(1, D-1). \tag{10}$$

Integration of its density $(D-1)(1-t)^{D-2}$ gives Eq. (9); its mean is $1/D$. □

Thus the typical squared overlap is of order D^{-1} and the typical amplitude is of order $D^{-1/2}$. The exact distribution is more informative than a generic application of Lévy's lemma because it retains the precise dependence on the overlap threshold.

B. A finite-family theorem

The quantity relevant for a measurement with several alternatives is the largest pairwise overlap, rather than the overlap of one pair.

Proposition 2 (Simultaneous overlap and trace-distance bound)**.** *Let $m \ge 2$, let $|E_1\rangle, \ldots, |E_m\rangle$ be independent Haar-random unit vectors in $\mathbb{C}^D$, and let $0 < \eta < 1$. Define*

$$K = \binom{m}{2}, \qquad x_\eta = \frac{\log(K/\eta)}{D-1}, \qquad \varepsilon_\eta = 1 - e^{-x_\eta}. \tag{11}$$

With probability at least $1-\eta$,

$$\max_{i\neq j} |\langle E_j|E_i\rangle| \le \sqrt{\varepsilon_\eta} \le \sqrt{\frac{\log[\binom{m}{2}/\eta]}{D-1}}. \tag{12}$$

Consequently, the branched state in Eq. (1) *obeys, with the same probability,*

$$D_{\mathrm{tr}}(\rho_S, \Delta(\rho_S)) \leq \min\left\{1, \frac{1}{2}\left[\left(\sum_i |c_i|\right)^2 - 1\right]\sqrt{\varepsilon_\eta}\right\}$$
$$\leq \min\left\{1, \frac{m-1}{2}\sqrt{\frac{\log[\binom{m}{2}/\eta]}{D-1}}\right\}. \quad (13)$$

Proof. Lemma 1 and a union bound give

$$\mathbb{P}\left(\max_{i<j} |\langle E_j|E_i\rangle|^2 > \varepsilon\right) \leq \binom{m}{2}(1-\varepsilon)^{D-1}. \quad (14)$$

Equation (11) makes the right-hand side equal to η. The second inequality in Eq. (12) follows from $1 - e^{-x} \leq x$. Proposition 1 completes the proof. □

For fixed m and fixed confidence, the operational error in this bound is $O(D^{-1/2})$. If m grows with D, the accumulation factor in Eq. (13) must be retained. This qualification is important in discussions of very large branch families. For comparable amplitudes the displayed bound scales as $m\sqrt{\log(m)/D}$ and therefore becomes nontrivial in a considerably more restricted regime than the pairwise packing bound below. The union bound does not require mutual independence of the pair-overlap events. The same conclusion holds for any joint ensemble whose pairwise marginals have the Haar overlap distribution of Lemma 1.

The tradeoff can also be read directly from Eq. (11). At fixed D and failure probability η, increasing the number of branches m increases the overlap threshold ε_η that can be guaranteed. Equivalently, keeping both η and the allowed overlap fixed while increasing m requires a larger effective dimension D.

C. Packing is not classical memory capacity

The same union-bound calculation gives a random-coding statement. For $0 \leq \varepsilon < 1$, let $M_\varepsilon(D)$ denote the largest cardinality of a family of unit vectors in $\mathbb{C}^D$ satisfying $|\langle\psi_i|\psi_j\rangle|^2 \leq \varepsilon$ for every $i \neq j$. Then

$$M_\varepsilon(D) \geq \left\lfloor \exp\left[\frac{D-1}{2}\big(-\log(1-\varepsilon)\big)\right]\right\rfloor. \quad (15)$$

Indeed, for this number of independent Haar samples the union-bound probability of a violating pair is strictly below one. For fixed small ε, Eq. (15) is exponential in εD.

This spherical-code packing bound is not a capacity for reliably readable classical records. When $M > D$, the states are linearly dependent and cannot be perfectly discriminated by a single-copy measurement. The Welch bound further gives

$$\max_{i\neq j} |\langle\psi_i|\psi_j\rangle|^2 \geq \frac{M-D}{D(M-1)} \qquad (M > D), \quad (16)$$

Equation (16) is a geometric restriction, not by itself a bound on a specified decoder [12]. Approximate discrimination requires a prior distribution, a measurement, and an error criterion. For a uniform ensemble of M pure states in dimension D, the Holevo bound $\chi \leq \log D$, combined with Fano's inequality, gives a nonzero lower bound on the average decoding error when $\log M - \log D$ is sufficiently large [13, 14]. Equation (15) is therefore best viewed as a statement about projective Hilbert-space geometry. The operationally relevant decoherence statement is instead Proposition 2 for a specified family of alternatives.

IV. MIXED ENVIRONMENTS: COHERENCE IS NOT A RECORD

This section isolates a counterintuitive but important distinction. Greater initial mixedness can reduce the Haar-averaged decoherence factor while simultaneously reducing, or even eliminating, the environment's capacity to encode a readable record. These are different operational quantities, not a paradox.

For an initially mixed environmental state ρ_E, a controlled unitary produces reduced-system coherence factors

$$\gamma_{ji}(t) = \mathrm{Tr}\big[U_i(t)\rho_E U_j(t)^\dagger\big] = \mathrm{Tr}[\rho_E W_{ji}(t)]\,,\; W_{ji}(t) = U_j(t)^\dagger U_i(t). \quad (17)$$

The relevant random object is the *relative* conditional unitary $W_{ji}(t)$, not two independently postulated purifications.

Lemma 2 (Haar relative unitary)**.** *Let W be Haar-random on a D-dimensional Hilbert space and let ρ be a fixed density operator, chosen independently of W, on that space. Then*

$$\mathbb{E}_W|\,\mathrm{Tr}(\rho W)|^2 = \frac{\mathrm{Tr}(\rho^2)}{D}. \quad (18)$$

Proof. In a fixed basis, Haar invariance gives the second-moment identity

$$\mathbb{E}_W[W_{ab}W^*_{cd}] = \frac{1}{D}\delta_{ac}\delta_{bd}. \quad (19)$$

Expanding $|\,\mathrm{Tr}(\rho W)|^2$ and applying this identity yields $D^{-1}\sum_{a,b}|\rho_{ab}|^2 = D^{-1}\,\mathrm{Tr}(\rho^2)$. □

The identity requires only the first unitary-design moment, equivalently the matrix-entry correlation $\mathbb{E}[W_{ab}W^*_{cd}]$. It immediately yields a simultaneous mixed-state statement.

Proposition 3 (Mixed-environment simultaneous bounds)**.** *Let ρ_E and the normalized coefficient vector (c_i) be fixed independently of a specified ensemble of relative unitaries. Suppose that for every $i < j$ the marginal ensemble of W_{ji} reproduces the matrix-entry* *correlation* $\mathbb{E}[W_{ab}W^*_{cd}] = D^{-1}\delta_{ac}\delta_{bd}$ *on a* D*-dimensional sector. The probability refers to the specified ensemble, such as Hamiltonian disorder, circuit randomness, conditional-perturbation randomness, or a prescribed random-time sampling procedure. No mutual independence of the different* W_{ji} *is required. Put* $K = \binom{m}{2}$ *and let* $0 < \eta < 1$*. Then*

$$\mathbb{P}\left\{\max_{i<j} |\gamma_{ji}| \geq \delta\right\} \leq K\,\frac{\mathrm{Tr}(\rho_E^2)}{D\delta^2}, \tag{20}$$

and hence, with probability at least $1-\eta$,

$$\max_{i<j} |\gamma_{ji}| \leq \min\left\{1, \sqrt{\frac{K\,\mathrm{Tr}(\rho_E^2)}{D\eta}}\right\}. \tag{21}$$

Moreover, with probability at least $1-\eta$,

$$D_{\mathrm{tr}}(\rho_S, \Delta(\rho_S)) \leq \min\left\{1, \frac{1}{2}\sqrt{\frac{m\,\mathrm{Tr}(\rho_E^2)}{D\eta}\left(1 - \sum_i |c_i|^4\right)}\right\}. \tag{22}$$

Proof. Markov's inequality and Lemma 2 give $\mathbb{P}\{|\gamma_{ji}| \geq \delta\} \leq \mathrm{Tr}(\rho_E^2)/(D\delta^2)$. A union bound over the K pairs proves Eqs. (20) and (21). For the stronger trace-distance estimate, define

$$Y = \sum_{i \neq j} |c_i|^2 |c_j|^2 |\gamma_{ji}|^2. \tag{23}$$

The assumed marginal second moments imply

$$\mathbb{E}Y = \frac{\mathrm{Tr}(\rho_E^2)}{D}\left(1 - \sum_i |c_i|^4\right). \tag{24}$$

Markov's inequality gives $Y \leq \mathbb{E}Y/\eta$ with probability at least $1-\eta$. Substitution into Eq. (5) proves Eq. (22). □

Writing $d_{\mathrm{purity}} = 1/\mathrm{Tr}(\rho_E^2)$ makes the mean-square scale

$$\frac{\mathrm{Tr}(\rho_E^2)}{D} = \frac{1}{D d_{\mathrm{purity}}}. \tag{25}$$

Thus D characterizes the relative-unitary ensemble, whereas d_{purity} characterizes the initial environmental mixedness.

For a pure environment, Eq. (18) reduces to $1/D$. For the maximally mixed state it gives $1/D^2$. This stronger suppression does *not* mean that a maximally mixed environment stores a better record. The conditional environmental states are

$$\rho_E^{(i)} = U_i \rho_E U_i^\dagger. \tag{26}$$

If $\rho_E = I/D$, all of them are identical, so no measurement on the environment can reveal i, even though $\mathrm{Tr}(W_{ji})/D$ may be small.

For pure conditional states, the connection between overlap and record distinguishability is direct. With equal priors, the minimum error probability for discriminating two pure states is

$$p_{\mathrm{err}} = \frac{1}{2}\left(1 - \sqrt{1 - |\langle E_j|E_i\rangle|^2}\right). \tag{27}$$

Equivalently,

$$D_{\mathrm{tr}}(|E_i\rangle\langle E_i|, |E_j\rangle\langle E_j|) = \sqrt{1 - |\langle E_j|E_i\rangle|^2}. \tag{28}$$

For mixed states, however, distinguishability is governed by trace distance or fidelity between the density operators in Eq. (26), not by a chosen purification overlap alone. In particular, unitary invariance gives

$$D_{\mathrm{tr}}\left(\rho_E^{(i)}, \rho_E^{(j)}\right) = \frac{1}{2}\left\|W_{ji}\rho_E W_{ji}^\dagger - \rho_E\right\|_1, \tag{29}$$

which is generally unrelated to the magnitude of $\mathrm{Tr}(\rho_E W_{ji})$. Decoherence, locally accessible records, and redundant records are therefore different properties. The latter requires information about the pointer variable to be recoverable from many disjoint environmental fragments [15–17]. Indeed, generic global random states need not have the branching structure needed for redundant storage [15]. For initially mixed environments or more general mixed global states, system decoherence can occur without system–environment entanglement or objective records [18]. By contrast, for an initially pure global state, nontrivial reduced-state mixedness entails system–environment entanglement.

V. THE DYNAMICAL BRIDGE

The preceding propositions are kinematic. In a microscopic pure-dephasing model with time-independent conditional environmental Hamiltonians H_i,

$$\gamma_{ji}(t) = \langle E_0|e^{iH_j t}e^{-iH_i t}|E_0\rangle. \tag{30}$$

Its modulus squared is a Loschmidt echo. Decoherence can consequently be studied using fidelity decay, spectral correlations, perturbation theory, and many-body orthogonality effects [19, 20]. Equation (30) also exposes why a large Hilbert space is insufficient: $e^{iH_j t}e^{-iH_i t}$ may remain close to the identity, preserve large invariant subspaces, or display revivals long before any Poincaré recurrence.

Several dimensions should be distinguished. For an energy window $[E, E+\Delta E]$, let

$$D_{\mathrm{shell}} = \mathrm{Tr}\ \mathbf{1}_{[E,E+\Delta E]}(H_E). \tag{31}$$

The Boltzmann entropy associated with that convention is

$$S_B(E, \Delta E) = k_B \log D_{\mathrm{shell}}. \tag{32}$$

By contrast, let $H_E = \sum_\lambda E_\lambda P_\lambda$ be the spectral decomposition into distinct energy eigenspaces and put

$$p_\lambda = \langle E_0|P_\lambda|E_0\rangle, \qquad d_{\rm spec} = \frac{1}{\sum_\lambda p_\lambda^2}. \tag{33}$$

This spectral participation ratio measures how broadly the initial state is distributed over distinct energy eigenspaces and is widely used in equilibration bounds [21–23]. For a nondegenerate Hamiltonian and $|E_0\rangle = \sum_\alpha a_\alpha |E_\alpha\rangle$, it reduces to $d_{\rm spec} = 1/\sum_\alpha |a_\alpha|^4$. The projector formulation avoids dependence on an arbitrary basis chosen inside a degenerate eigenspace. Neither $D_{\rm shell}$ nor $d_{\rm spec}$ is automatically the dimension D appearing in a Haar model for $W_{ji}(t)$.

For chaotic many-body systems, eigenstate thermalization and spectral statistics offer reasons to expect equilibration of appropriate observables, but they do not imply that a single finite-time orbit is Haar-random over an entire shell [24, 25]. Conserved charges, locality, finite propagation speed, integrability, many-body localization, and weak conditional perturbations may all obstruct the Haar model. A defensible application therefore requires a statement about the ensemble or time distribution of the relative unitaries $W_{ji}(t)$. At a fixed time, fixed Hamiltonians and a fixed initial state determine $W_{ji}(t)$ deterministically. Every probability statement must therefore identify its probability source, such as Hamiltonian disorder, a random circuit ensemble, random initial states, a distribution of conditional perturbations, or a random time sampled from a specified window. A temporal distribution along one orbit agrees with an ensemble distribution only under an additional ergodicity or sampling assumption and generally contains correlated samples. For time-dependent $H_i(t)$, the exponentials in Eq. (30) must be replaced by the appropriate time-ordered propagators.

Approximate unitary designs provide one possible intermediate assumption. The mean-square identity in Lemma 2 requires only the relevant second moment, whereas a tail comparable to Eq. (9) requires stronger moment or concentration control. Local random circuits are known to approach approximate designs only after a depth that depends on locality, system size, design order, and accuracy [26]. This makes circuit depth or physical evolution time a meaningful replacement for an unqualified appeal to Haar typicality.

VI. A LOCAL PRODUCT MECHANISM

Many decoherence models have a more physical route to small overlaps than global Haar randomness. As an idealized mechanism, suppose that the environment is initially factorized, residual interactions between fragments can be neglected on the timescale of interest, and the conditional dynamics preserves the product form

$$|E_i\rangle = \bigotimes_{k=1}^{N} |e_i^{(k)}\rangle. \tag{34}$$

Consequently,

$$\begin{aligned} \langle E_j|E_i\rangle &= \prod_{k=1}^{N} \langle e_j^{(k)}|e_i^{(k)}\rangle, \\ -\log|\langle E_j|E_i\rangle| &= \sum_{k=1}^{N} -\log|\langle e_j^{(k)}|e_i^{(k)}\rangle|. \end{aligned} \tag{35}$$

The logarithmic identity assumes nonzero local overlaps; if one local overlap vanishes, the global overlap vanishes already. If a positive density of fragments has local overlaps bounded above by a fixed $q < 1$, or more generally if

$$\liminf_{N\to\infty} \frac{1}{N} \sum_{k=1}^{N} \Big[-\log|\langle e_j^{(k)}|e_i^{(k)}\rangle|\Big] > 0, \tag{36}$$

Eq. (35) gives exponential suppression in N without requiring the global state to be Haar-random. At the same time, the fragment structure makes the question of redundant records well posed. This local mechanism and the global geometric estimate are complementary, not interchangeable. Approximate rather than exact conditional factorization produces model-dependent correction terms.

VII. THERMODYNAMIC-LIMIT COMPARISON: INFINITE TENSOR PRODUCTS AND SECTORIZATION

The finite-dimensional estimates above concern generic Haar-distributed pairs, whose overlaps are nonzero almost surely but typically very small. Finite-dimensional Hilbert spaces can of course contain exactly orthogonal states, and a finite product overlap is exactly zero if even one corresponding pair of local factors is orthogonal. A related but conceptually distinct phenomenon occurs when macroscopically different reference sequences define disjoint representation sectors in an infinite tensor-product limit.

Here N denotes the number of tensor factors, whereas D in the preceding sections denotes the dimension of the sector on which the random-state or relative-unitary model is imposed; in many-body applications D may itself grow exponentially with N. Consider two sequences of normalized local states,

$$|E_i^{(N)}\rangle = \bigotimes_{k=1}^{N} |e_i^{(k)}\rangle, \qquad |E_j^{(N)}\rangle = \bigotimes_{k=1}^{N} |e_j^{(k)}\rangle. \tag{37}$$

Their overlap factorizes as

$$\langle E_j^{(N)}|E_i^{(N)}\rangle = \prod_{k=1}^{N} \langle e_j^{(k)}|e_i^{(k)}\rangle. \tag{38}$$

Let n_N be the number of factors among the first N for which

$$\Big|\langle e_j^{(k)}|e_i^{(k)}\rangle\Big| \le q < 1, \tag{39}$$

and suppose that $n_N/N \to \alpha > 0$. Then the overlap vanishes exponentially,

$$\begin{aligned}\left|\langle E_j^{(N)}|E_i^{(N)}\rangle\right| &\le q^{n_N} = \exp[-n_N|\log q|] \\ &= \exp[-\alpha N|\log q| + o(N)] \longrightarrow 0,\end{aligned} \quad (40)$$

Thus the sequence of finite-product overlaps tends to zero in the thermodynamic limit. This limiting statement alone does not yet establish inequivalent representations: an infinite tensor-product representation and an algebra of quasi-local observables must also be specified.

Von Neumann's construction of infinite tensor products makes this limit more precise [27]. For two normalized reference sequences, a standard sufficient criterion for their failure to belong to the same weak equivalence class is

$$\sum_{k=1}^{\infty}\left(1 - |\langle e_j^{(k)}|e_i^{(k)}\rangle|\right) = \infty. \quad (41)$$

The positive-density condition above implies this divergence. In the standard infinite-product construction, such reference sequences can define orthogonal incomplete tensor-product sectors and, under the corresponding quasi-local formulation, disjoint or unitarily inequivalent representations of the observable algebra. Here "unitarily inequivalent" concerns representations of the physical observable algebra, not the absence of an abstract isomorphism between Hilbert spaces of the same dimension.

This distinction is important. At every finite N, all product states are vectors in the same finite-dimensional Hilbert space and can be related by finite-system unitaries. Different thermodynamic-limit states may instead define disjoint representations of the quasi-local algebra. The latter conclusion is representation dependent and does not follow from finite-N geometry alone. It is one mathematical framework used in thermodynamic-limit approaches to measurement theory [28–30].

For infinitely many nontrivial finite-dimensional factors, the complete infinite tensor product is generally nonseparable and decomposes into mutually orthogonal sectors, while an individual incomplete tensor-product sector can remain separable. The complete product may carry continuum many sector labels. Writing this cardinal specifically as $\aleph_1$ assumes the continuum hypothesis. The physically relevant distinction is the emergence of disjoint representations of the quasi-local observable algebra, not the cardinality label by itself.

This infinite-system mechanism may therefore be viewed as an idealized exact counterpart of the finite-dimensional geometry developed above: finite environments give exponentially small overlaps, whereas a specified infinite-product construction can yield disjoint sectors. It does not, by itself, establish that a physical environment is literally infinite, select one sector as the unique realized outcome, or derive the Born probabilities. It does permit macroscopically distinct phases or outcomes to be represented in inequivalent sectors rather than forcing them into a single irreducible representation.

VIII. SCOPE OF THE GEOMETRIC STATEMENT

The geometric calculation establishes the following conditional implication: if a specified finite family of environmental branches is distributed like independent typical vectors in a D-dimensional sector, then its largest pairwise overlap and the operational distance of the reduced system from its dephased state are small with high probability. It also shows that the projective Hilbert space contains exponentially large pairwise quasi-orthogonal codes.

It does not establish any of the following claims. First, it does not select a pointer basis; this depends on the structure and timescales of the system–environment interaction. Second, it does not prove that a physical Hamiltonian generates Haar-typical relative branches. Third, it does not turn an improper mixture into a proper ensemble or select a unique outcome. Fourth, it does not equate small system coherence with a readable, let alone redundant, environmental record. Finally, exponential spherical packing does not provide exponentially many perfectly decodable classical labels.

The operational content relevant to Schrödinger's "jellification" worry is therefore limited but precise. For every POVM effect $0 \le M \le I$,

$$\left|\mathrm{Tr}\big[M\big(\rho_S - \Delta(\rho_S)\big)\big]\right| \le D_{\mathrm{tr}}(\rho_S, \Delta(\rho_S)). \quad (42)$$

The supremum of the left-hand side over all effects equals the trace distance. For a Hermitian observable A with $\|A\|_\infty \le 1$, the corresponding expectation-value difference is bounded by $2D_{\mathrm{tr}}$. Propositions 2 and 3 therefore directly bound changes in system measurement probabilities under their respective typicality assumptions. They do not assert that the global coherent superposition has vanished.

IX. FURTHER RESEARCH

A first concrete test is to replace the Haar assumption by conditional local Hamiltonians $H_i = H_E + V_i$. One can compare chaotic, integrable, and localized spin environments by measuring the decay, long-time plateau, temporal variance, and full distribution of Eq. (30). Histograms of $|\gamma_{ij}(t)|^2$ can be compared directly with the beta law in Lemma 1, but the sampling prescription must be stated. An ensemble histogram and a time histogram along one correlated orbit are not interchangeable without an ergodicity and sampling analysis. Their deviation from the beta law is itself a measure of the failure of pair typicality.

A second problem is finite-time design formation. For conditional random circuits or noisy Floquet systems, one can seek explicit bounds of the form

$$\mathbb{P}\{|\operatorname{Tr}(\rho_E W_{ji}(t))|^2 \geq \varepsilon\} \leq f(D, t, \varepsilon, \operatorname{Tr}\rho_E^2), \tag{43}$$

interpolating between short-time perturbative behavior and a random-matrix plateau. Here $\mathbb{P}$ must refer to a specified circuit, disorder, initial-state, perturbation, or random-time ensemble. Locality should appear explicitly through circuit depth or a Lieb–Robinson scale.

A third direction is the many-branch problem. Rather than controlling only the largest entry, one can study the spectrum of the Gram matrix $G_{ij} = \langle E_j | E_i \rangle$ and derive bounds tailored to the coefficient vector (c_i). This would sharpen Eq. (4) and identify when numerous weak coherences collectively remain observable.

Finally, for mixed environments one should analyze two quantities in parallel: the decoherence factor $\operatorname{Tr}(\rho_E W_{ji})$ and the distinguishability or accessible information of the conditional fragment states $\rho_F^{(i)}$. Their separation is necessary for determining when geometric decoherence is accompanied by objective, redundantly accessible records.

X. CONCLUSION

High-dimensional Hilbert-space geometry supplies a useful but conditional component of decoherence. Independent Haar-random pure states have the exact overlap tail $(1-\varepsilon)^{D-1}$. For a finite family, this tail yields a high-probability bound on both the largest branch overlap and the trace distance between the reduced system and its pointer-basis dephasing. For a mixed initial environment, Haar-random relative dynamics instead gives the mean-square coherence $\operatorname{Tr}(\rho_E^2)/D$ and the simultaneous bounds of Proposition 3. The Hilbert–Schmidt estimate (5) additionally controls the collective contribution of many weak coherences. It provides a complementary collective bound and can be substantially sharper than the entrywise estimate when many weak coherences are broadly distributed, although it is not uniformly sharper.

These results quantify capacity, not dynamics. The physical question is whether conditional relative unitaries generated by a particular local Hamiltonian acquire the necessary typicality on the relevant timescale. Moreover, loss of system coherence, environmental distinguishability, and redundant record formation remain separate notions. With these distinctions made explicit, high-dimensional geometry provides a clean quantitative bound on locally accessible interference without being asked to solve the preferred basis, definite-outcome, or objectivity problems by itself.

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